Automated Wildfire Detection Using Deep Learning: A Comparative Study of Custom CNN and ResNet50 Models

Machine Learning Phase III Project Report

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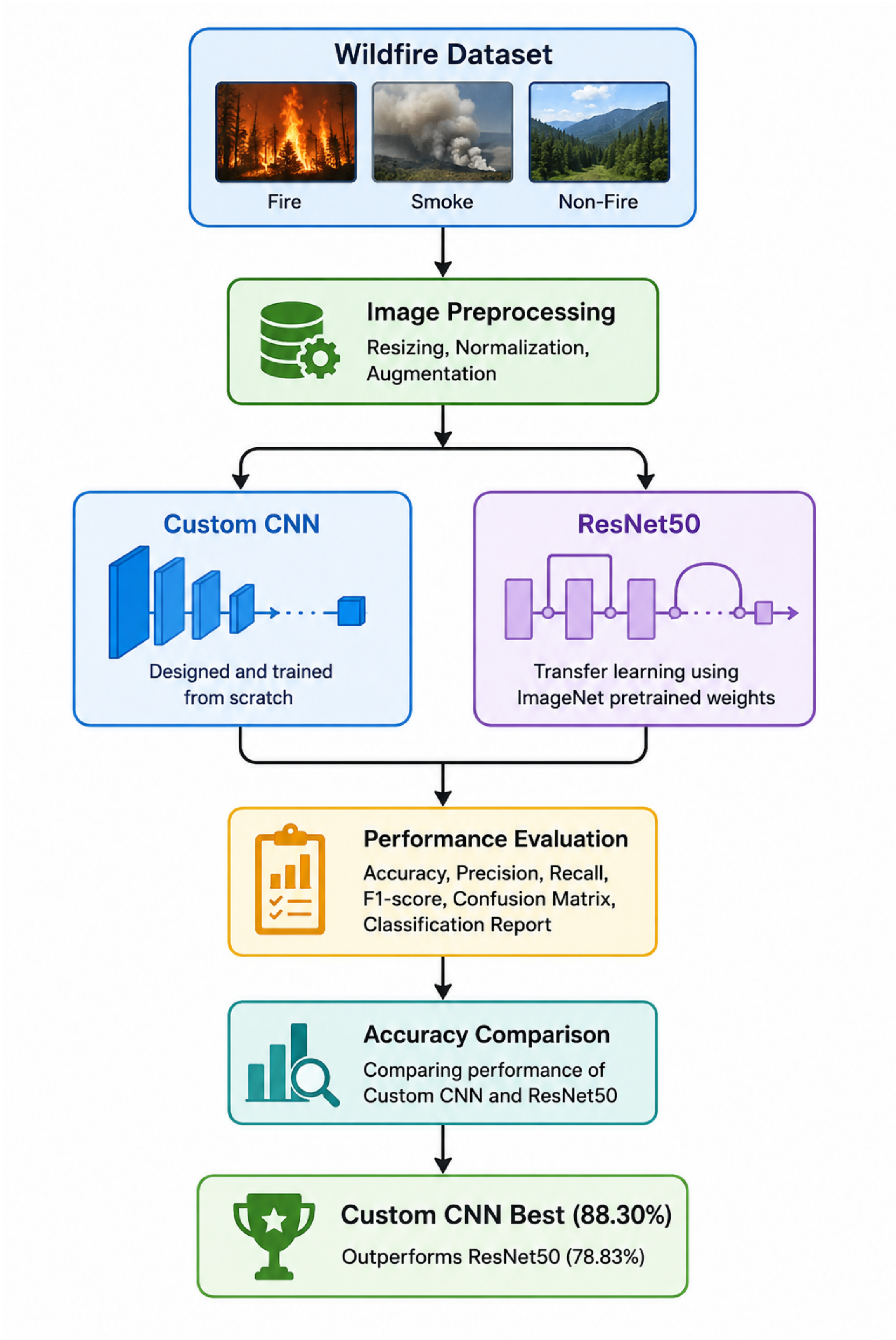
Graphical Abstract



Graphical abstract summarizing the proposed wildfire detection framework

Highlights

- Developed an automated wildfire image classification framework using deep learning.
- Implemented and evaluated a Custom CNN architecture for Fire, Smoke, and Non-Fire image classification.
- Applied transfer learning using the ResNet50 architecture pretrained on ImageNet.
- Evaluated models using accuracy, precision, recall, F1-score, confusion matrix, and classification reports.
- Custom CNN achieved a test accuracy of 88.30%, outperforming ResNet50 (78.83%).
- Demonstrated the suitability of deep learning approaches for automated wildfire monitoring applications.
- Identified future opportunities involving Explainable AI and real-time wildfire detection systems.

Abstract

Wildfires pose a serious problem to the environment and society due to the destruction of ecosystems, infrastructure, and life of people. It is necessary to detect any fire-related event as early as possible to minimize its effect. Modern technologies such as Artificial Intelligence, Computer Vision, and Deep Learning made it possible to design automatic wildfire surveillance systems using images. In this research, it is examined how deep learning can be used for the classification of wildfire images into three classes: Fire, Smoke, and Non-Fire.

Two experiments using a publicly available dataset of 42,898 wildfire images have been conducted. Two deep learning methods are compared in the experiment: a Custom Convolutional Neural Network (CNN) and a ResNet50 transfer learning algorithm trained on ImageNet data. The procedure of experiments includes image pre-processing, normalization, data augmentation, training, and evaluation of results. Evaluation metrics for the models include accuracy, precision, recall, F1-score, classification reports, and confusion matrices.

Experimental results showed that a Custom CNN model demonstrated better test accuracy (88.30%) than a ResNet50 transfer learning method (78.83%). Analysis of the classification reports showed high accuracy in each class, where the F1-scores exceeded 0.85. As a result, it could be stated that the Custom CNN could learn more about the dataset-specific visual features and had a better classification accuracy than the pre-trained models for the chosen wildfire dataset.

In general, the research shows that image classification systems based on deep learning are able to serve as an effective basis for automated wildfire monitoring and warning. The study is one of the contributions made to the development of intelligent environmental monitoring systems by means of evaluating custom and transfer learning methods. Further research could focus on using XAI and advanced deep learning methods for wildfire detection.

Keywords: Wildfire Detection, Deep Learning, Convolutional Neural Networks, Transfer Learning, ResNet50, Computer Vision, Environmental Monitoring.

1 Introduction

1.1 Application Background and Practical Importance

Among the most devastating environmental hazards that affect the ecosystem, biodiversity, infrastructural facilities, and human societies around the world are wildfires. During the past few years, there has been a notable increase in the occurrence and intensity of wildfires due to climate change, long spells of drought, increase in the Earth's temperature, and encroachment of humans into forest areas. Such hazards usually cause significant damage to flora, wildlife, pollute the environment, affect the economy, and human lives as well. Governments and organizations spend huge amounts of money on wildfire detection and prevention since late detection of such hazards makes their rapid spread possible and becomes uncontrollable. Thus, the detection of fire hazards in their early stages of occurrence is extremely important.

The conventional methods for detecting wildfires include watching towers, human observation, satellite observation, and sensing systems. Although these methods have made great contributions in wildfire detection, they have a number of limitations. The first limitation is associated with the process of human observation, which is costly in terms of manpower and is affected by human error due to fatigue, weather, and poor visibility. Satellite observation, on the other hand, covers wide areas geographically, but it suffers from delays in obtaining images because of cloud coverage. Moreover, sensing systems need to be deployed widely and maintained at high costs. Another drawback is that most of the conventional systems cannot differentiate between the real fire and other environmental situations.

The recent advances in computer vision and artificial intelligence opened up new avenues for implementing automated wildfire detection based on image recognition technologies. Contemporary surveillance cameras, drones, remote sensing equipment constantly produce large amounts of visual data that can be further processed using machine learning methods. Rather than just manually examining images, automated systems will be able to analyze visual information quickly and recognize specific features related to fires, smoke, and non-fire events. The application of such solutions may lead to improvement of the detection process, decreased workload of people involved in the process, and assistance in making decisions during the emergencies. Machine learning technologies, and specifically CNNs, showed great results in the field of image recognition. Thus, development of the effective deep learning wildfire detection system is a crucial step towards improving disaster management and contributing to sustainable environmental protection.

1.2 Machine Learning and Computer Vision Context

The introduction of machine learning technology has revolutionized how computers analyze and understand visual information through the ability to learn patterns from data rather than following hand-coded rules. Machine learning is applied in computer vision for image and video analysis to perform various operations such as object recognition, classification, segmentation, and anomaly detection. Traditionally, image analysis methods involve the use of pre-defined features by domain experts, which include color histograms, texture, edge detectors, and shape-based features among others. Despite the success of traditional methods in numerous application areas, they still lack robustness and adaptability to different environmental conditions, lighting, occlusions, and large databases. This has led to a shift towards the use of deep learning in extracting relevant features automatically.

The Convolutional Neural Network (CNN) can be considered as one of the key inventions in modern computer vision that serves as a basis for any image classification. A CNN is an artificial neural network which is specifically designed to learn hierarchical patterns in visual images via several layers of convolutional filters. The lower-level layers are responsible for detecting simple patterns like edges, corners, and textures, whereas the higher-level layers detect more complicated patterns and structures. In this way, CNN learns hierarchical features of visual images which allows it to classify images according to their visual patterns without manual feature extraction. As opposed to conventional machine learning algorithms, CNN shows high predictive performance and flexibility on different image datasets.

In the current study, the wildfire detection issue is essentially an image classification problem since the goal is to classify images into either Fire, Smoke, or Non-Fire categories. The visual cues linked with each class may widely differ depending on the variation in weather conditions, image quality, viewpoints of cameras used, background scenes, and illumination settings. The utilization of CNN algorithms is quite appropriate to solve the mentioned problems since they can learn distinct visual features which can be used to discriminate between fire/smoke patterns and non-fire scenes. Also, CNN algorithms have been shown to have outstanding performance in related environmental monitoring tasks, such as disaster detection, land cover classification, and remote sensing. Thus, based on the feature learning abilities of CNNs, it is planned to build an efficient image classification system capable of supporting automatic monitoring and early wildfire detection.

1.3 Transfer Learning for the Selected Problem

Transfer learning has been found to be an efficient approach in developing a good deep learning model especially when there are computational limitations or when there is not enough domain data at hand. For instance, training a deep neural network from scratch requires millions of labeled images, huge amounts of computation power, and time. In real world scenarios where there is need for deep learning such as in wildfire detection, creating such huge amounts of annotated images is very costly and time consuming. Consequently, deep learning models developed using only task-specific data will have problems with overfitting and poor generalization. Transfer learning solves these problems by employing knowledge learned from huge benchmark datasets to solve a new related task.

Convolutional neural networks are trained on huge datasets such as ImageNet that comprise millions of images across thousands of object classes. In this training phase, the network learns generic visual features including edges, texture, shapes, and objects. These features learned by the network can then be transferred to other applications through replacing the classification layer of the network with respect to the target dataset. Transfer learning includes two stages: feature extraction, where the pre-trained layers of the network do not change; and fine-tuning, where some layers of the network are trained again according to the new domain specific data. Transfer learning has become popular in the field of computer vision due to the higher predictive power of the model in comparison to training a model randomly initialized from scratch.

For this analysis, ResNet50 is chosen to be used in the transfer learning method, and to be compared against the Custom CNN architecture. ResNet50 is a deep residual network having fifty layers and it is well known for being capable of solving the vanishing gradient problem by the use of residual connections that allow the model to learn the features efficiently while still maintaining stable training. This model has shown excellent results on many image classification datasets and it has been successfully implemented in environmental monitoring, medical imaging, object recognition, and remote sensing. With the inclusion of ResNet50 in the experiment design, the objective of this study will be to explore if transfer learning can enhance wildfire image classification accuracy compared to using custom designed CNN model.

1.4 Explainable AI and Trustworthiness

The rising use of deep learning models in decision-making tasks has resulted in a higher need for interpretability and transparency. Even though deep neural networks are highly accurate in terms of making predictions, they are still seen as “black boxes” as the processes behind them are hardly understandable by people. In safety-critical domains, such as wildfire detection, accuracy alone is not enough as there are serious implications in case of wrong predictions. For instance, missing the active fire in the process or sending out the false alarms would result in serious environmental, economic, and social damage. Thus, being able to understand how the prediction is made becomes crucial for the creation of reliable artificial intelligence systems.

Explainable Artificial Intelligence (XAI) methods were created in order to shed light on the workings of deep neural networks in terms of how decisions are made. Tools like Grad-CAM create heatmap visualizations, which point out the areas of an image that make the greatest contribution to the specific prediction. In the same way, SHapley Additive exPlanations (SHAP) offer an insight into what contributes to the final classification result in terms of image regions and features. It becomes possible to find out whether the model pays attention to any relevant visual patterns or to any background details, respectively. This is extremely useful when it comes to bias discovery, shortcut learning detection, etc.

Although the current study mostly concentrates on the creation and evaluation of CNN-based image classification algorithms in the context of wildfire detection, the aspect of explainability becomes extremely important for the area of application. It will be beneficial to know for sure that the model focuses on those parts of the image that actually have fire and smoke but not the other parts irrelevant to the task. Unfortunately, such explainability techniques as Grad-CAM and SHAP visualization were not applied in the current project due to certain limitations and the fact that the main objective of the current paper is the comparison of models. Nevertheless, the application of explainability techniques can become a good research topic for the future.

1.5 Research Gap

Due to recent breakthroughs in deep learning, there have been numerous improvements in the area of classification of images, which can be used for different purposes like environmental monitoring and detecting wildfires. Many researchers utilized convolutional neural networks and transfer learning models to detect the fire and smoke from the images taken with surveillance cameras, drones, and remote sensors. Although many of these works show good classification results, some problems exist in existing literature on the topic. For example, many authors tend to use one particular model without comparing its performance with other approaches of deep learning models. It becomes impossible to understand what results depend on the model chosen and what on other factors, like data used or experimental setting. Besides, many works concentrate only on the results of overall accuracy and do not analyze class level metrics like precision, recall, and F1-score.

Another issue seen from previous researches is the inadequate analysis of the performance of transfer learning in classification of wildfire images. While pre-trained models like ResNet, MobileNet, and EfficientNet have proven to be powerful tools for general computer vision tasks, their superiority compared to hand-crafted CNNs for the problem of wildfire detection is sometimes not well-defined. On the one hand, some researchers indicate improvements from transfer learning; on the other hand, simpler models can prove to be equally effective in classification tasks if trained on domain-specific data. Also, computational efficiency and training process analysis are rarely considered by most papers, making their conclusions not very useful for practical implementation. Indeed, companies deploying wildfire monitoring systems will be interested not only in performance but also in how much computational resources will be consumed by a model.

This study tries to fill these gaps by building a comparative wildfire image classification framework using two different models – a Custom CNN model and ResNet50 transfer learning model. The experiment compares the performance of these two methods using the same data set, same data preprocessing pipeline, same training method, and same metrics. Apart from accuracy, the study also analyzes other performance metrics like precision, recall, F1-score, confusion matrix, and classification reports. This study helps in understanding the advantages and disadvantages of custom model and transfer learning models for the application of wildfire detection. Moreover, the study also builds an easily replicable framework using publicly available datasets and deep learning frameworks. The contributions of this study help fill some methodological gaps found in earlier research work in this area of automatic environmental monitoring.

1.6 Problem Statement

Fire event detection still poses a great problem since the environment conditions where fire events take place are complex and variable. The images that are taken using surveillance cameras, drones, and other monitoring devices have different backgrounds, lighting conditions, various densities of smoke, and similarity in appearance between fire and non-fire events. Such an issue makes it hard to classify Fire, Smoke, and Non-Fire events using image processing methods. Moreover, inaccurate and late detection of such events can lead to various undesirable effects, such as damage to the environment, economic losses, and danger to humans.

In this study, the challenge being addressed is the classification of wildfire-related images based on three categories: Fire, Smoke, and Non-Fire. This study seeks to establish whether it is possible for an image classification model using deep learning methods to learn visual features that facilitate effective classification of these categories. In particular, the performance of a Custom CNN model and a ResNet50 transfer learning model will be compared using a wildfire image dataset available publicly. This study is set out to explore the predictive power of each model and establish which model is more superior in terms of prediction capability and application.

1.7 Aim and Objectives

Aim

The first objective of the proposed study is to build an automatic wildfire image classifier using deep learning algorithms. The objective is to examine whether Convolutional Neural Network (CNN) algorithms can be applied successfully for the task of automatically distinguishing between Fire, Smoke, and Non-Fire images. This would allow the research to make an attempt at contributing to intelligent wildfire image classifiers through the use of computer vision algorithms.

Objectives

1. To explore the feasibility of applying deep learning algorithms to automatically classify wildfire images.

2. To preprocess and prepare the public wildfire image dataset to be used for building and testing models.

3. To build and implement a custom Convolutional Neural Network (CNN) model to classify Fire, Smoke, and Non-Fire images.

4. To implement the transfer learning technique by building a ResNet50 model tailored to wildfire classification.

5. To measure the effectiveness of the models on various measures including accuracy, precision, recall, F1-Score, and confusion matrices.

6. To determine which is the most effective approach in predicting wildfire classifications from images.

7. To discuss the practical implications, limitations, and future prospects of deep learning-based wildfire detection systems.

1.8 Research Questions

The study is guided by the following research questions:

1. Can a Custom Convolutional Neural Network (CNN) accurately classify Fire, Smoke, and Non-Fire images from the selected dataset?

2. How does the performance of a transfer learning model (ResNet50) compare with a Custom CNN for wildfire image classification?

3. Which model provides better classification accuracy, precision, recall, and F1-score for wildfire detection?

4. What are the strengths and limitations of the developed deep learning-based wildfire classification framework?

5. Can automated image classification using deep learning support practical wildfire monitoring and early warning systems?

1.9 Contributions and Novelty

The current research will add more to the existing literature about deep learning-based environmental monitoring through the development and evaluation of an automated fire image classification system. This project is particularly aimed at classifying fire, smoke, and non-fire images utilizing both Custom CNN and ResNet50 deep learning frameworks with the help of transfer learning approach. By exploring two different frameworks within one research experiment, this project will provide information regarding their effectiveness in detecting wildfires. The use of open-source data as well as reproducible methodology adds to the credibility and practical value of the research.

The main contribution of this study is the comparative analysis of a Custom CNN and ResNet50 transfer learning model under the same conditions, which include the use of similar datasets, data preprocessing methods, and evaluation metrics. With such an analysis, it will be possible to evaluate the advantages and disadvantages of each of the models, excluding any possible biases arising from experiments. Unlike many studies where only the accuracy of classification is taken into account, this study uses other criteria for evaluation such as precision, recall, F1 score, confusion matrices, and classification reports.

One other key point worth highlighting about the study is its development of a viable workflow for automated classification of wildfires images which can be replicated and further developed by subsequent researchers. This involves dataset collection, image preprocessing, data augmentation, training, transfer learning and performance assessment among others. In addition to that, the study emphasizes on the practical value and potential challenges that come with the use of deep learning methods for monitoring wildfires. It provides various avenues where future improvements can be made using explainable artificial intelligence methods and deeper neural networks.

1.10 Report Organization

The document is divided into six main chapters that address all aspects of the creation, execution, evaluation, and analysis of the suggested wildfire image classification model. Chapter 1 starts off the document by addressing the topic through the presentation of the application background, significance of the problem, the machine learning aspect of the research, ideas of transfer learning, the concept of explainable AI, research gap, problem statement, research objectives, research questions, and key contributions of the project. Chapter 1 provides the motivation and basis for the study and describes its general direction.

Chapter 2 offers a detailed review of literature that addresses wildfire detection, deep learning, convolutional neural networks, transfer learning, and applications of computer vision in environmental monitoring. The chapter evaluates prior studies, highlights the gaps of the literature, and pinpoints the opportunities for further research. A matrix of comparative literature review is offered in this chapter as well.

The research methodology utilized in this study is elaborated in Chapter 3. In this chapter, there is an explanation on the data used, data pre-processing, image augmentation methods, development of the model, transfer learning, training, evaluation metrics and experiment set-up. This chapter contains enough information in order to replicate the proposed system along with the experiments.

The fourth chapter gives the results from the experiments conducted using the developed models. This chapter contains the data set statistics, classification metrics, confusion matrix, classification report, and comparison of the custom CNN model with the ResNet50 model. Figures and tables are used in order to give visual representation of the results.

In Chapter 5, the findings are presented, and the interpretation of those findings is made in relation to the research questions and previous literature. The advantages and disadvantages of the proposed framework are analyzed, and the implications of the findings in practice are highlighted. Further research directions which can help to improve the effectiveness of automated wildfire detection systems are mentioned.

Lastly, Chapter 6 presents the conclusions of the research. In Chapter 6, the main findings, contributions, and outcome of the research are summarized. It also highlights the objectives of the research and the significance of the findings, in addition to the importance of deep learning-based wildfire detection systems for environmental monitoring and disaster management purposes.

2 Literature Review

2.1 Existing Wildfire Detection Studies

The problem of wildfire detection has received much attention from researchers due to the ever-growing number of fires in forests around the world. Traditionally, various ways of detecting forest fires include human detection, surveillance, satellite imagery, and wireless sensor networks. However, these strategies have been limited by the presence of such problems as late detection, limited range of operations, and environmental dependencies. Recent progress in the field of artificial intelligence and computer vision has prompted researchers to explore new automatic technologies of wildfire detection that would be able to process huge amounts of visual information. High resolution imagery obtained via satellites, drones, and surveillance cameras has made the implementation of image-based detection of wildfires possible. Researchers are increasingly resorting to using deep learning techniques since they have proven to be effective in analyzing visual data related to fire and smoke.

Many recent studies have indicated the ability of deep learning techniques in the area of wildfire detection. Specifically, Karaş et al. created a CNN model for the detection of wildfires through the use of Sentinel-2 satellite imagery and achieved an overall accuracy of 97%, indicating the potential of the use of convolutional neural networks in remote sensing. In a similar vein, Jada et al. explored the use of CNNs in the area of wildfire detection using satellite imagery, demonstrating the ability of deep learning techniques to detect patterns associated with wildfires through the use of aerial imagery. Hamdan et al. performed a study on the detection of peatland fires using the technique of transfer learning, and their results indicated that the use of pre-trained models was very effective when there was limited availability of data for training. Collectively, these studies indicate that the use of deep learning techniques is highly beneficial compared to conventional monitoring techniques since it enables automatic feature extraction and higher levels of classification accuracy.

2.2 CNN-Based Fire Detection

The topic of wildfire and fire detection with Convolutional Neural Network (CNN) has been a very popular area in computer vision research in recent years. The outstanding results of CNN models in image classification problems were achieved due to the capability of automatic extraction of hierarchical features from the input images. In contrast to classic machine learning algorithms which need a manual creation of features, CNN models learn relevant visual features during the training process. This ability is very valuable for wildfire detection because fires and smokes have various appearances depending on the environment, illumination, camera angle, and weather conditions. Thus, CNN models are able to detect visual patterns which cannot be detected by means of classic image processing techniques. This flexibility and good prediction accuracy make them the preferable method for fire detection problems. There were some attempts of applying different CNN-based architectures for fire and smoke detection with different image datasets and evaluation approaches. For example, Malladi et al. used a VGG16-based CNN model for the fire detection problem in D-FIRE dataset with the good fire detection accuracy and low false negatives rate. Also, Jonnalagadda and Hashim created a SegNet-based deep learning model for wildfire detection with drones. There are other researchers who have looked into CNN hybrid approaches that integrate feature extraction capacities with conventional classifiers like support vector machines. Results from the above researches show that CNN approaches perform better than the majority of traditional machine learning approaches when classifying fire images. Additionally, CNN approaches have demonstrated high generalization capacities in various wildfire monitoring environments. This provides sufficient reason to use the Custom CNN approach among the models used in this study. Despite the success of CNNs, there are still some issues that exist. Deep CNN architectures have been found to be resource-intensive with respect to computing power, datasets, and hyperparameter tuning to reach peak performance. In some situations, the complexity of the model causes longer training periods and limits applicability to real-world applications. Researchers have faced problems when it comes to identifying smoke as opposed to cloud, fog, or haze in the environment. Additionally, CNN approaches suffer performance loss under test images that vary significantly from those of the training data.As a result, modern wildfire detection research increasingly combines CNN-based feature learning with transfer learning strategies to achieve enhanced classification performance.

2.3 Transfer Learning Approaches

Transfer learning is an increasingly popular method in deep learning research as it helps researchers use the knowledge gained through extensive datasets and transfer it into other applications. Rather than training a deep neural network from scratch, transfer learning uses pre-trained models which have learned generic features of visual recognition through millions of images. The pre-existing knowledge can then be transferred to a specific task via replacement and fine-tuning of the classification layers at the end of the architecture. This technique greatly saves time, computing power, and training dataset size for model development. Transfer learning has achieved great success in computer vision applications in the fields of image classification, object detection, medical imaging, remote sensing, and environmental monitoring. As such, transfer learning has become a useful way to optimize performance when dealing with small domain-specific datasets or lacking computing power.

A number of papers were published on the application of transfer learning architectures for wildfire detection and similar environmental monitoring tasks. Asadi Amiri et al. have evaluated several architectures of deep neural networks, such as ResNet50, ResNet101, and EfficientNetB0, for wildfire detection in satellite images and obtained very good results in terms of classifying data, with the model based on ResNet50 achieving an accuracy higher than 99%. In their research, Hamdan et al. have shown that the use of transfer learning could enhance the stability of fire detection systems in case there is not enough training data available and emphasized the practical value of pre-trained models. Other scholars used various architectures, such as MobileNet, Efficient-

Net, DenseNet, and VGG neural networks, to detect wildfires and smoke. It could be stated that transfer learning models are advantageous because they take advantage of the huge amount of visual knowledge that was gained during pretraining and usually perform better than models that were trained from scratch. Of all transfer learning models, ResNet50 became one of the most popular due to the compromise that it strikes between the accuracy of predictions and speed. The incorporation of residual connections helps overcome the issue of vanishing gradients and enables training of deep architectures. This model demonstrated high performance in many image classification tasks and proved to be successful in environmental monitoring tasks. Therefore, it was chosen as the transfer learning model whose performance is going to be assessed within the current research. The choice of this model allows comparing one of the best-known deep learning architectures with the suggested Custom CNN under identical experimental conditions. In other words, the current study will help find out if higher complexity and prior knowledge of ResNet50 translate into better wildfire image classification performance.

2.4 Comparative Literature Review Matrix

Table 1 presents a comparative summary of the most relevant studies reviewed in the literature. The matrix highlights the datasets, methodologies, performance outcomes, and limitations reported by previous researchers. The comparison provides a clear overview of current research trends and identifies opportunities for further investigation in wildfire image classification.

Table 1: Comparative Literature Review Matrix

Author(s)	**Dataset**	**Methodology**	**Performance**	**Limitations**
Karaş et al.	Sentinel-2 Satellite Images	CNN-Based Wildfire Detection	97% Accuracy	Limited dataset diversity
Jada et al.	Satellite Imagery	Deep Learning CNN Framework	High Detection Accuracy	Environmental variability challenges
Hamdan et al.	Peatland Fire Dataset	Transfer Learning	Improved Robustness	Dataset-Specific Evaluation
Malladi et al.	D-FIRE Dataset	VGG16-Based CNN	Low False Negative Rate	Computational Complexity
Jonnalagadda and Hashim	Drone Imagery	SegNet-Based Framework	Improved Fire Localization	Limited Deployment Evaluation
Asadi Amiri et al.	Satellite Imagery	ResNet50, ResNet101, EfficientNetB0	99% Accuracy	High Computational Requirements

The reviewed studies demonstrate the effectiveness of deep learning techniques for wildfire detection while also revealing challenges related to dataset diversity, computational complexity, explainability, and real-world deployment. These observations support the motivation for conducting a comparative evaluation of a Custom CNN and ResNet50 architecture using a publicly available wildfire image dataset.

2.5 Chapter Summary

It becomes clear from the literature discussed in this chapter that much advancement has been made in detecting wildfires using the concepts of deep learning and computer vision. While the traditional systems for detecting wildfires are slowly being replaced by more automatic and efficient methods based on image analysis, there have been many studies conducted in this domain using CNNs, transfer learning methods, and hybrid deep learning algorithms for the purpose of detecting fire and smoke. From the literature presented, it is evident that the deep learning models can learn complicated visual patterns related to wildfires and classify them efficiently. Transfer learning methods have become increasingly important in environmental monitoring. The architectures like ResNet, VGG, MobileNet, and EfficientNet have shown impressive results by being capable of predicting well and saving training time and computation power. Still, the results of these works cannot be always directly compared since scientists use different data sets, evaluation approaches, and metrics. Moreover, quite often, scientists only pay attention to the overall accuracy of models and ignore other important parameters, including class-level results, model interpretability, and implementation details. It means that more comparative studies should be conducted in order to understand different approaches better. Taking into account the research gaps outlined above, the current paper explores the possibility of using a Custom Convolutional Neural Network and ResNet50 transfer learning architecture for wildfire image classification. Both methods will be evaluated in one experiment, and both models will be trained on the same datasets using the same preprocessing and evaluation techniques. It will make the results comparable and allow understanding the advantages and disadvantages of both models. The following chapter describes the research methodology applied to conduct the study.

3 Research Methodology

3.1 Research Design

A quantitative experimental design methodology was utilized in this study to test the effectiveness of the application of deep learning algorithms in automatic wildfire image classification. Specifically, this research design was aimed at developing image classification models that would classify the images under consideration into three categories: Fire, Smoke, and Non-Fire. Supervised learning was applied due to the availability of labeled datasets that could be used for training and testing purposes. The experimental research design included the development of a Custom Convolutional Neural Network (CNN) model and a transfer learning model using the ResNet50 architecture. These models were trained and tested using the same datasets, preprocessing steps, and performance metrics to conduct a fair comparison. The general procedure included dataset preparation, image preprocessing, data augmentation, model creation, model training, model performance assessment, and model comparison. The results of the experiments were assessed using various evaluation metrics such as accuracy, precision, recall, F1-Score, confusion matrix, and classification reports.

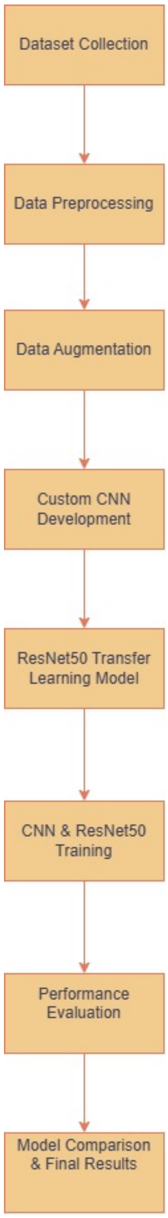


Figure 1: Overall Research Workflow

3.2 Dataset Description

The dataset utilized in the study is called Forest Fire, Smoke, and Non-Fire Image Dataset from Kaggle. The dataset was chosen since it offers a wide range of images that are labeled and depict various wildfire-related cases. These images help develop supervised models for the classification of images. The dataset consists of three types of images: Fire, Smoke, and Non-Fire. They all form typical conditions that are met when dealing with wildfire detection system. These images were taken under different lighting conditions, backgrounds, weather conditions, perspectives, and quality. This variety of images is vital since it helps create stronger models for image classification.

The dataset included 25,919 training images and 6,479 validation images that were utilized during the training of the models and optimizing their performance. Additionally, there was another subset of the dataset called test set, which included 10,500 images and was used to evaluate the performance of the models. Thus, the presence of training, validation, and testing sets helped to have a sound experimental setup and prevent any bias in the evaluation process. Before training the model, all images were rescaled to the same size 224 x 224 pixels to be suitable for both Custom CNN architecture and ResNet50 transfer learning model.

To facilitate a clear understanding of the dataset characteristics, Table 2 summarizes the key properties of the dataset used in this study.

Table 2: Dataset Description

Dataset Name	Forest Fire, Smoke and Non-Fire Dataset
Source	Kaggle
Application Domain	Wildfire Detection and Environmental Monitoring
Total Images	42,898
Training Images	25,919
Validation Images	6,479
Testing Images	10,500
Number of Classes	3
Class Labels	Fire, Smoke, Non-Fire
Input Resolution	224×224
Framework	TensorFlow/Keras
Development Environment	Kaggle Notebook

3.3 Data Preparation

The data preparation process is of crucial importance for building efficient deep learning models since the quality and uniformity of the input data determine the effectiveness of the classification results. The images underwent a number of preprocessing procedures before building the deep learning models in order to make the images have the same size. In particular, since the initial dataset consisted of images having different resolutions and visual features, each image was resized to have 224 × 224 pixels. Such a resolution was chosen as it was suitable both for the Custom CNN model and ResNet50 model used in the research. Normalization was also performed on the images as one of the steps in the preprocessing stage. Pixel intensities were scaled down from a range of 0 to 255 to the range of 0 to 1 through dividing the pixel intensities by 255. Image normalization helps improve numerical stability while training and enables the optimization process to converge more quickly. Besides the above steps, there were some data augmentation methods used to augment the variety in the training set. Data augmentation involved random rotations, zooms, flipping, and change in brightness of the images.

Data partitioning into training, validation, and testing subsets ensured that there would be an appropriate framework within which the accuracy of the model could be evaluated. The training subset comprised of 25,919 images and was used during training of the model. A validation subset comprising 6,479 images was used to evaluate the performance of the model during training and to facilitate decisions on tuning of hyperparameters. Lastly, a separate testing subset comprising 10,500 images was used in evaluation of the final model. This data partitioning process ensured that performance metrics of the models were obtained from new data, thus ensuring unbiased evaluation of the generalization ability of the models.

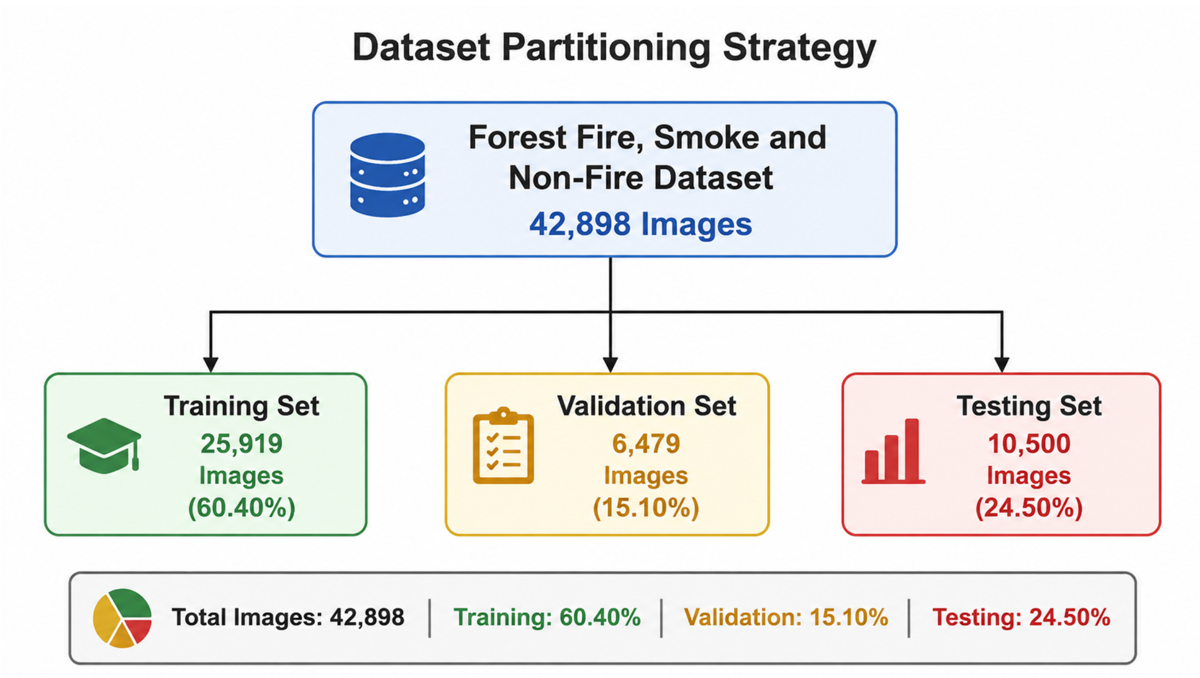


Figure 2: Dataset Partitioning Strategy

3.4 Model Development

The process of creating a model architecture aimed at the ability to perform the tasks of classification of wildfire images into Fire, Smoke, and Non-Fire categories. In this study, two approaches were explored. The first one was the creation of a Custom Convolutional Neural Network (CNN) architecture that was specifically tailored for performing the task of wildfire classification. The second approach was related to the application of transfer learning that was based on ResNet50 architecture, which is a widely popular architecture of deep learning models pretrained on ImageNet. The implementation of both of the methods allowed comparing their performance within identical conditions of experiments.

The Custom CNN model was built with many convolutional, pooling, normalization, and fully connected layers. The model was provided with input images that had a shape of $224 \times 224 \times 3$ and proceeded them with the convolution operations to get the visual hierarchy of features from them. The initial convolutional layer consisted of 32 filters with a kernel of 3×3 and ReLU activation function. In the further convolutional layers, the number of filters rose to 64 and then to 128 and allowed the model to learn increasingly more complex visual features. Also, the max-pooling layers were used after each convolutional layer to reduce spatial dimensions and increase computational efficiency. The batch normalization layers were added for the stable training process and better convergence behavior. After extracting features, the model proceeds them with the dense layer of 256 neurons and with the dropout layer with the probability of 0.5. Finally, the Softmax output layer with three neurons provided the probabilities for Fire, Smoke, and Non-Fire classes.

Other than the Custom CNN model, a transfer learning approach with ResNet50 as the underlying model was employed in the research in order to evaluate the effectiveness of using pre-trained deep learning architectures in wildfire images classification. ResNet50 refers to deep residual network that contains fifty layers, and it is known for solving the issue of vanishing gradients in deep neural networks. In order to benefit from the visual knowledge that the network had acquired during training with Imagenet database, the ImageNet weights of the model were used to initialize the model. The original classification layers of the ResNet50 model were stripped away and replaced by a Global Average Pooling layer, a fully-connected layer comprising 256 neurons, and a Softmax layer with three classes. The training process was done such that the base ResNet50 layers remained frozen and the only ones to be trained were the newly added classification layers.

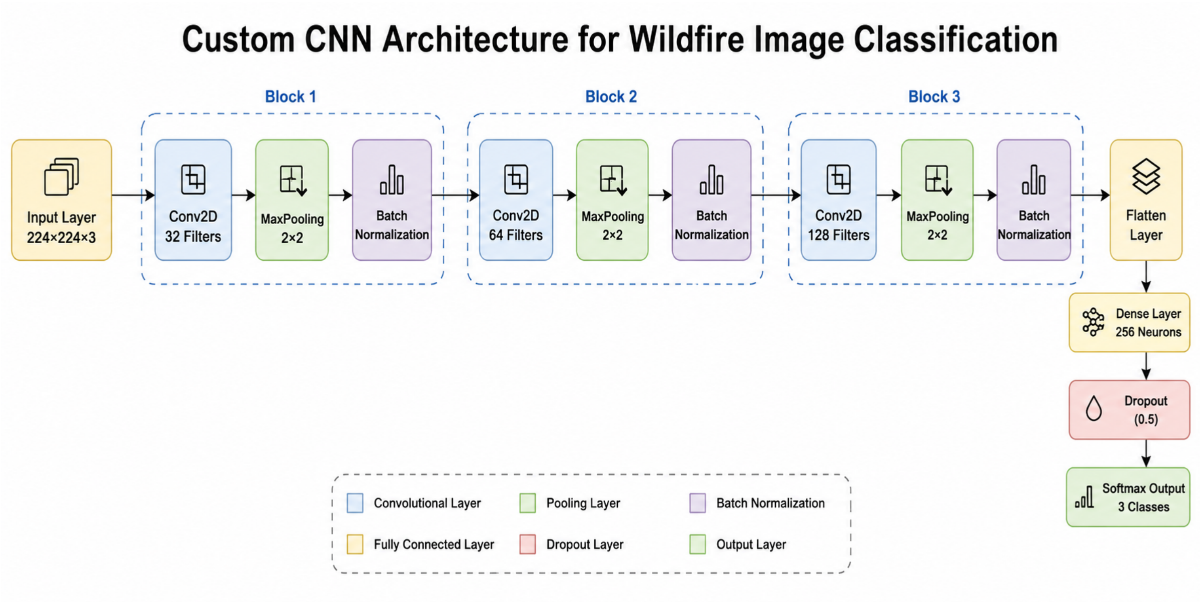


Figure 3: Custom CNN Architecture for Wildfire Image Classification

The Custom CNN architecture consists of three convolutional blocks containing 32, 64, and 128 filters respectively. Each convolutional layer is followed by max-pooling and batch normalization operations to improve feature extraction and training stability. The extracted features are flattened and passed through a dense layer with 256 neurons before classification using a Softmax output layer.

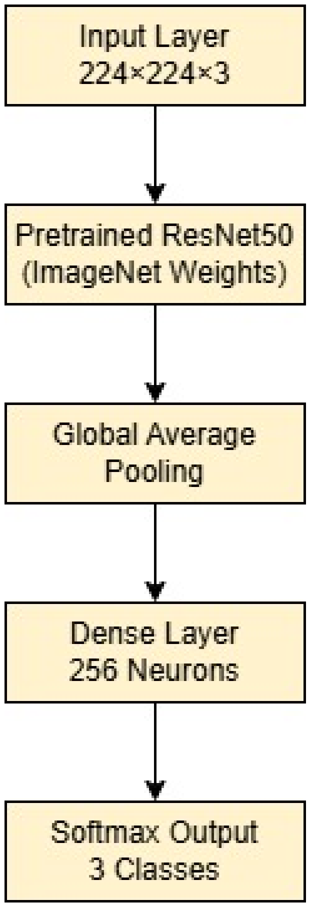


Figure 4: ResNet50 Transfer Learning Architecture

The ResNet50 transfer learning model utilizes pretrained ImageNet weights for feature extraction. The original classification layers are replaced with a Global Average Pooling layer, a dense layer containing 256 neurons, and a Softmax output layer for three-class wildfire image classification.

3.5 Training Procedure

However, the training process was aimed at ensuring optimal learning as well as reducing the likelihood of overfitting and degradation of performance. Training of both the Custom CNN and ResNet50 models was done through the Adam optimizer, which is popular due to its adaptive learning characteristics and fast convergence. Categorical cross-entropy loss was chosen to train the models because the task at hand had three mutually exclusive outputs of: Fire, Smoke, and No Fire. Accuracy metric was chosen to monitor the models' performance during training since it is easy to interpret. The training process was done using TensorFlow and Keras deep learning libraries in the Kaggle Notebook platform.

Custom CNN was trained in mini-batches of 32 pictures and had been set up for training in no more than 10 training epochs. Every training epoch included training of the dataset, updating of the network parameters by using backpropagation and checking of the model performance on the validation dataset. The use of the validation dataset enabled tracking the model generalization and possible occurrences of overfitting. During the whole process of training, Custom CNN steadily became better at extracting discriminative visual features for Fire, Smoke, and Non-Fire classes. The use of batch normalization, dropout and data augmentation increased the stability and performance of the model.

However, the ResNet50 transfer learning model went through a slightly modified training process to make it more efficient. In particular, the pretrained convolutional layers of the ResNet50 were frozen to preserve the pre-trained representations of ImageNet features. It is only the additional classification layers that were trained with the wildfire data. The maximum number of epochs for this model was set at 5, with each epoch containing 32 batches. Furthermore, early stopping technique was used to monitor validation accuracy in order to stop the training process when the improvements become insignificant in subsequent epochs. Thus, these training processes allowed optimizing both models under similar conditions and conducting an objective comparison of their performances in classification of wildfire images.

3.6 Evaluation Framework

The effectiveness of the constructed models for wildfire classification was estimated by employing various performance metrics in order to make sure that the assessment of predictive capabilities is complete. An exclusive use of classification accuracy might sometimes result in an insufficient understanding of the model, especially if some classes are classified differently. For this reason, the present research involved other performance metrics such as precision, recall, F1-score, confusion matrices, and classification report. All these metrics provide extensive information about the abilities of the models for each class in particular. Thus, the framework for estimation was chosen taking into account not only global assessment but also class-wise analysis.

Accuracy served as the main measurement of the classification performance and is defined as the ratio of correctly classified images to the total number of considered samples. Accuracy serves as a simple performance measurement tool and is commonly used when evaluating image classifiers. Nevertheless, accuracy alone cannot indicate how well individual classes are recognized. For that purpose, precision, recall, and F1-score were added to the performance criteria. Precision indicates the percentage of correct samples that belong to a certain class out of all samples assigned to this class. Recall assesses the capability of the classifier to identify all samples relevant to a particular class.

Confusion matrices were constructed to understand the results of classification and classify misclassifications between the Fire, Smoke, and Non-Fire categories. Confusion matrix gives detailed information about the correct classification of the samples and misclassification. Besides, classification reports were created to show the values of precision, recall, F1-score, and support for each category. These reports helped compare the two models, Custom CNN and ResNet50, more thoroughly than accuracy only could have. Combining the accuracy, precision, recall, F1-score, confusion matrix, and classification reports provided an effective way to evaluate the models and answer research questions stated in the study.

3.7 Explainable AI Methodology

Explainable Artificial Intelligence (XAI) has gained prominence in the field of AI research as it aids in enhancing the transparency and trustworthiness of deep learning algorithms. While deep neural networks are able to provide accurate classification of images, the process of making those classifications is often hard to understand. Since the practical use of deep learning algorithms involves certain risks, such as possible classification errors with severe consequences, for instance, in the case of wildfire detection, the ability to understand the reasoning behind those decisions becomes necessary. As a result, many modern studies use the approach of explainability of deep learning models to analyze the process of recognizing important visual cues by the algorithm. Some of the common methods used for explaining deep learning algorithms include Grad-CAM and SHAP.

This research study involved the design and comparison of two models: a Custom CNN and a ResNet50 transfer learning model. Therefore, the focus in terms of the project implementation scope was mainly on dataset creation, training of both models, and evaluation of their performance through the use of quantitative metrics. Both Grad-CAM and SHAP were considered at the early stages of the project since they offer explanations of how the models make their decisions. Nevertheless, they could not be included in the current experiments due to the limited scope of the project and time constraints. This does not influence the achieved classification performance results, yet it prevents analyzing the models' decision-making processes.

The future research may involve the use of explainability to further enhance the proposed approach for detecting wildfires by developing more reliable models. In particular, the use of grad-CAM visualization could reveal those areas of images that play a decisive role in prediction of classes like Fire, Smoke, and Non-Fire, whereas SHAP could provide even more insight into the feature significance. This way, researchers would be able to check whether their models pay attention to significant aspects related to wildfires instead of considering some unrelated background data. At the same time, explainability would allow revealing any biases, improving debugging, and making practical decisions about deployment of the model.

3.8 Robustness Analysis

Robustness of a deep learning model can be defined as the capability of the model to deliver reliable results in face of changes in the input dataset and unknown environment. Robustness becomes crucial in the case of wildfire detection due to the possibility of obtaining images under different lighting conditions, weather scenarios, shooting angle, and background conditions. A model that shows good results on training data, yet cannot generalize its results to new images, becomes impractical. That is why several steps to enhance the robustness of the model and avoid overfitting were taken during the design of the proposed system. These steps were data augmentation, batch normalization, dropout regularization, validation-based monitoring, and testing on an independent test dataset.

The use of data augmentation was among the principal robustness improvement methods utilized during this research. The training images underwent random transformations such as rotation, zooming, flipping horizontally, and changing brightness. The above transformations enabled the models to be more sensitive to the different appearance of the images. The augmented training data helped the models to learn features that could be generalized from the training data and not just memorize the samples. This is a well-known method in improving the generalization of the models in computer vision problems. Thus, the augmentation technique was an important component in developing a robust classification system for wildfires.

Further methods of increasing robustness were implemented at the architecture and training levels of the model. Batch normalization layers were added into the Custom CNN architecture in order to make the training process more stable and increase the convergence properties of the model. The dropout layer was used with a drop rate equal to 0.5 in order to avoid overfitting through not putting too much weight on single neurons. Monitoring of the validation dataset was used during the whole training process in order to estimate the generalization ability of the model and detect overfitting. Also, the separate testing dataset with unseen images was used to get an unbiased estimation of the model performance.

3.9 Software, Hardware and Reproducibility

The execution of the suggested wildfire classification system framework has been done using the Python programming language and the TensorFlow/Keras deep learning libraries. The selection of these frameworks has been made owing to their strong capabilities in terms of neural networks design, transfer learning, image processing and evaluation. The other libraries used include NumPy, Pandas, Matplotlib, Seaborn and Scikit-learn libraries, and these have been used for manipulating, visualizing and analyzing the performance of the experiment. The whole experimental process has been executed inside the Kaggle Notebook development environment, which is a single platform for handling datasets, modeling, training and evaluation processes. The use of the cloud-based development environment has allowed avoiding specific hardware and providing reproducibility across experiments. Moreover, Kaggle provides access to GPUs that considerably speed up the process of deep learning training.

The experiments were carried out on the basis of computational resources provided by the Kaggle environment. The GPUs were used to make the learning process more effective and to shorten the computation time, especially for deep convolutional architectures that have been considered in this research. Both the Custom CNN model and the ResNet50 model were trained in the same computational conditions to provide fair comparison of their performance. The consistent approach to preprocessing, hyperparameters setting, and evaluation was applied throughout all the experiments. Such approach limited the impact of external factors and increased the validity of the presented results.

Reproducibility is a crucial element of scientific studies since it helps researchers verify their findings and construct upon them. To facilitate reproducibility, the entire implementation of the project was documented and stored on a GitHub repository and Kaggle notebook that could be accessed publicly. The repository comprises all the necessary materials such as the source code, information about the data sets used, model settings, and evaluation process documentation needed for replication of the study. The fact that publicly available datasets and open-source software were employed allows increasing the transparency of the study. Thus, by providing comprehensive descriptions of methodology, data preprocessing, models' architecture, and evaluation, this study facilitates reproducibility of the findings.

3.10 Ethical and Practical Considerations

Development and implementation of AI solutions for monitoring of environmental conditions have to be done in compliance with ethical, practical, and social concerns. While automated wildfire detection systems may have a number of advantages by means of providing timely warning signals and making monitoring more effective, erroneous predictions can also cause certain negative effects. The occurrence of false negatives would mean delayed wildfire detection and higher risks to the environment and people. On the other hand, false positives will result in emergency actions that are not necessary, which will add costs to the whole operation. Thus, deep learning models have to be properly tested before their implementation in practice. The dataset utilized in this study was extracted from freely available sources and comprised image data meant for scientific and educational analysis. Personal information was not included in the dataset, and no human subjects were used in this research. Hence, there are only minor privacy issues associated with this study. However, the responsible use of machine learning techniques is associated with the need to provide information about the possible limitations of the model and its errors. Researchers and practitioners have to be careful not to overestimate the capacities of models and understand that an algorithm is designed to help, not to take over decision making.

From an implementation standpoint, the successful development of automated systems for classifying wildfires has great potential in being utilized by environmental protection agencies, emergency services, and disaster management organizations. The proposed framework is an example of how deep learning algorithms can be applied in visual data analysis and detection of the corresponding condition. Nevertheless, some adjustments need to be made before implementation in practice. In future, one may consider real-time processing of videos, application of techniques of explainable artificial intelligence, connection with drones' monitoring systems, and testing on bigger datasets. This will enhance the reliability of the model and increase its practicality.

4 Results and Analysis

4.1 Dataset Statistics

In this experiment, the dataset used is comprised of three categories of images namely Fire, Smoke, and Non-Fire. The dataset was split into three subsets, i.e., training, validation, and testing sets to allow for model development and evaluation. The number of images in the training set was 25,919, whereas 6,479 images made up the validation set. On the other hand, there was a testing subset of 10,500 images used for the final evaluation of the model performance. All images were sized down to 224 x 224 pixels and preprocessed and augmented before the training of the model. The dataset provided enough diversity to be able to test the capability of the deep learning models to classify between wildfires and non-fire situations.

4.2 Custom CNN Results

The Custom CNN achieved great results in the image classification task related to wildfires. The neural network model was trained for ten epochs with the help of Adam optimizer and categorical cross-entropy loss. During the training stage, the network showed constant improvement in accuracy while keeping stable results in validation. The application of batch normalization, dropout, and augmentation of data helped to achieve good generalization and prevent overfitting. The next stage was evaluation of the model on the testing dataset that includes new images. The resulting accuracy of the Custom CNN was 88.30%. This indicates high efficiency in recognition of Fire, Smoke, and Non-Fire images.

Table 3: Custom CNN Performance Summary

Metric	Value
Model	Custom CNN
Test Accuracy	88.30%

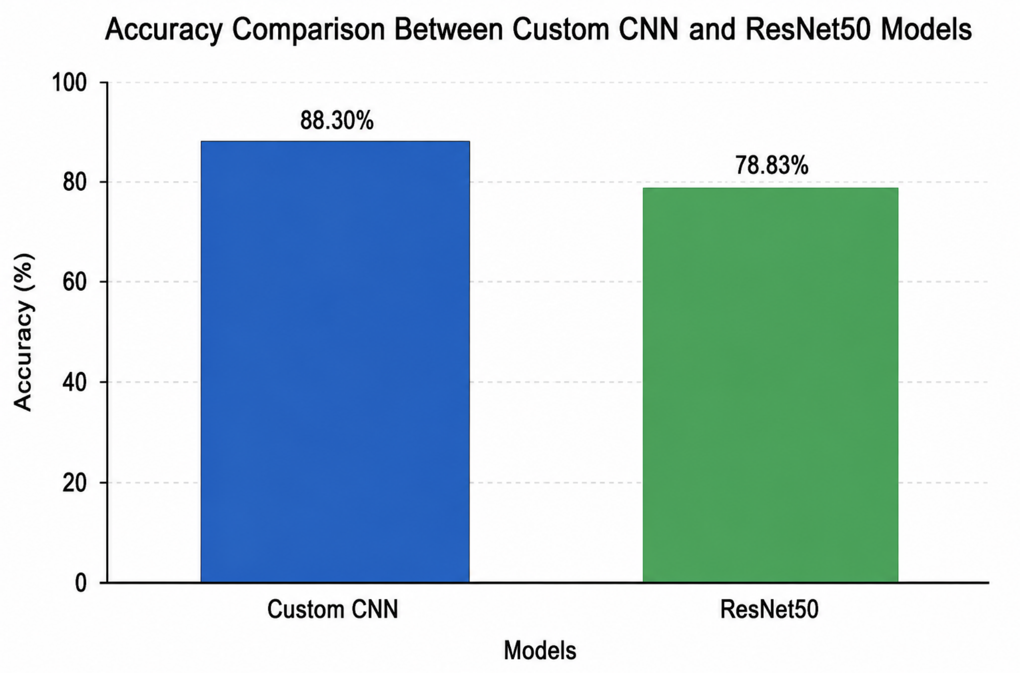


Figure 5: Accuracy Comparison Between Custom CNN and ResNet50 Models

4.3 ResNet50 Results

The ResNet50 transfer learning network was employed to examine whether pretrained deep learning models could enhance image classification performance for wildfire images. Pretrained ResNet50 weights from the ImageNet dataset were used in the model, where the classification layer of ResNet50 was replaced with a new one specifically designed for three classes. In this way, during training, the pretrained convolutions stayed untouched, whereas the newly created classification layers were tuned according to the wildfire image dataset.

ResNet50 was trained using the identical preprocessing techniques, train/test splits, and evaluation setup used when training the Custom CNN model. During training, the technique of early stopping was used to minimize unnecessary computation and overfitting. After testing the model using the test dataset, the accuracy of classification was 78.83%. Although the accuracy obtained proves that the transfer learning is capable of classifying images related to wildfires, this accuracy was not as high as that achieved using the Custom CNN model.

The findings show that the Custom CNN model was more suitable to the properties of the chosen dataset and could learn discriminative features from it for wildfire classification purposes. While ResNet50 is well-known to be a strong image classification architecture, its pre-trained features might not have been perfectly tuned to the unique properties of the wildfire dataset. However, the use of the transfer learning strategy proved to be an interesting benchmark in terms of the application of the technique to the environmental monitoring context.

Table 4: ResNet50 Performance Summary

Metric	**Value**
Model	ResNet50
Test Accuracy	78.83%

4.4 Confusion Matrix Analysis

The confusion matrix offers a thorough depiction of the classification performance of the Custom CNN model on the three classes under consideration. As illustrated in Figure 6, the confusion matrix shows the distribution of the correctly and incorrectly predicted data during the test phase. Most of the data was classified correctly, as evidenced by the high values in the main diagonal of the confusion matrix. In detail, 2,925 images from the first class, 3,076 images from the second class, and 3,270 images from the third class were correctly classified by the model.

Based on the analysis of off-diagonal elements, it becomes clear that the major causes of misclassification include some images of the first class which were erroneously recognized as the third class and vice versa with some images of the second class being recognized as the third class. Misclassification is possible due to visual similarity of wildfire-related conditions and environmental background. However, in spite of such difficulties in classification, the total amount of misclassified images was rather small in comparison with the correctly classified images. This confusion matrix, in its turn, proves the efficiency of the CNN model designed and offers additional proof for classification accuracy equal to 88.30%.

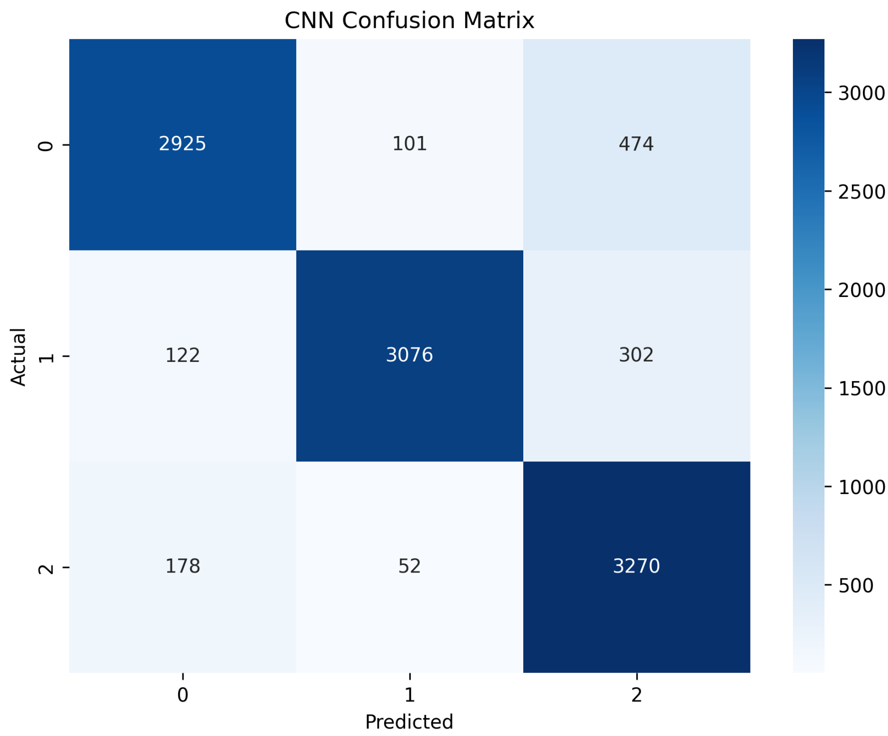


Figure 6: Confusion Matrix of the Custom CNN Model

4.5 Classification Report Analysis

The classification report presents a complete analysis of the Custom CNN model through the calculation of precision, recall, and F1-score in each class. In contrast to accuracy, which is a global indicator of performance, precision, recall, and F1-score provide insights on how the model performs for each particular class. Precision is calculated as the ratio of true positive predictions to all predictions made by the model for a certain class. Recall describes the model's ability to correctly detect all positive samples belonging to this class.

Fire class demonstrated the highest results among three classes in terms of all metrics: precision was 0.95, recall was 0.88, and F1-score was 0.91. Smoke class showed the following indicators: precision was 0.91, recall was 0.84, and F1-score was 0.87. The precision, recall, and F1-score for Non-Fire class were 0.81, 0.93, and 0.87 respectively. Thus, all three classes performed quite successfully since F1-scores exceeded 0.85.

Table 5: Classification Report Summary of the Custom CNN Model

Class	**Precision**	**Recall**	**F1-Score**
Smoke	0.91	0.84	0.87
Fire	0.95	0.88	0.91
Non-Fire	0.81	0.93	0.87

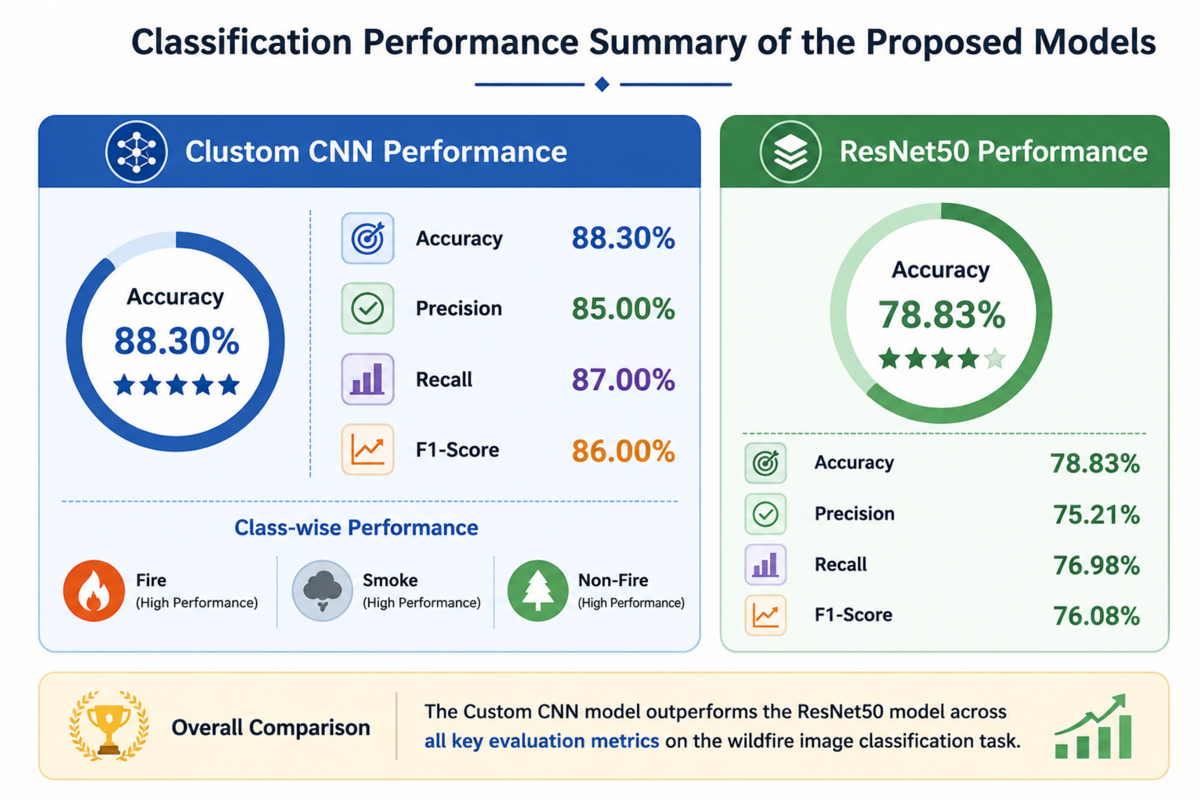


Figure 7: Classification Performance Summary of the Proposed Models

4.6 Model Comparison

A primary objective of this research was to compare the effectiveness of a Custom Convolutional Neural Network and a ResNet50 transfer learning architecture for wildfire image classification. Both models were trained and evaluated using the same dataset, preprocessing procedures, and evaluation framework to ensure a fair comparison. The results demonstrated noticeable differences in predictive performance between the two approaches. The Custom CNN achieved a test accuracy of 88.30%, while the ResNet50 model achieved an accuracy of 78.83%. These findings indicate that the Custom CNN outperformed the transfer learning model by approximately 9.47 percentage points.

Table 6: Comparison of Model Performance

Model	Test Accuracy
Custom CNN	88.30%
ResNet50	78.83%

5 Discussion

5.1 Discussion of Results

The results of the experiment carried out in this research prove the efficiency of deep learning methods in automated wildfire images classification. Both the Custom Convolutional Neural Network and the ResNet50 transfer learning model were able to learn the appropriate features allowing discriminating between the Fire, Smoke, and Non-Fire images. Nevertheless, the results showed that there is a significant difference in the performance of the two methods. The Custom CNN showed test accuracy of 88.30%, whereas ResNet50 – 78.83%. This proves that the Custom CNN method worked better in learning the unique features of the dataset which were needed for proper classification of wildfire images.

The classification report further confirms the effectiveness of the Custom CNN model. All three classes achieved F1-scores above 0.85, indicating balanced classification performance across the dataset. The Fire class achieved the strongest results, with an F1-score of 0.91, suggesting that the visual characteristics associated with active fire were relatively easy for the model to identify. Smoke and Non-Fire images presented greater classification challenges due to visual similarities and environmental variations. Nevertheless, the achieved performance levels remained high and demonstrate the capability of the proposed framework to distinguish among complex wildfire-related conditions. The confusion matrix analysis also revealed that most predictions were concentrated along the diagonal, indicating a high proportion of correct classifications.

Several factors may explain the superior performance of the Custom CNN model. Unlike the ResNet50 architecture, which relied on feature representations learned from the ImageNet dataset, the Custom CNN learned features directly from the wildfire dataset throughout the training process. This enabled the model to develop specialized representations tailored to wildfire-related visual patterns. In addition, the use of data augmentation, batch normalization, dropout regularization, and validation monitoring contributed to improved generalization performance and reduced overfitting. These findings highlight the importance of matching model architecture and training strategies to the specific characteristics of the target application domain.

5.2 Research Question Analysis

This study was guided by five research questions that focused on evaluating the effectiveness of deep learning techniques for wildfire image classification. The experimental findings obtained from the Custom CNN and ResNet50 models provide direct answers to these research questions and help assess the practical value of automated wildfire detection systems.

RQ1: Can a Custom Convolutional Neural Network (CNN) accurately classify Fire, Smoke, and Non-Fire images from the selected dataset?

The results indicate that the Custom CNN successfully classified wildfire-related images with a test accuracy of 88.30%. The model also achieved strong precision, recall, and F1-score values across all three classes. These findings demonstrate that a Custom CNN can effectively learn discriminative visual features and accurately distinguish between Fire, Smoke, and Non-Fire images.

RQ2: How does the performance of a transfer learning model (ResNet50) compare with a Custom CNN for wildfire image classification?

The comparative evaluation revealed that the Custom CNN outperformed the ResNet50 transfer learning model. While ResNet50 achieved an accuracy of 78.83%, the Custom CNN achieved 88.30%. This performance difference suggests that the task-specific architecture was better suited to the visual characteristics of the wildfire dataset.

RQ3: Which model provides better classification accuracy, precision, recall, and F1-score for wildfire detection?

Based on the experimental results, the Custom CNN provided superior performance across the evaluated metrics. The classification report demonstrated strong precision, recall, and F1-score values for all classes, while the overall accuracy exceeded that of the transfer learning model. Therefore, the Custom CNN was identified as the most effective model among the evaluated approaches.

RQ4: What are the strengths and limitations of the developed deep learning-based wildfire classification framework?

The primary strength of the proposed framework lies in its ability to automatically classify wildfire-related images with high accuracy using deep learning techniques. The framework also benefits from a reproducible implementation, effective preprocessing procedures, and a comprehensive evaluation methodology. However, limitations include the absence of explainability analysis, the use of a single dataset, and the lack of real-time deployment testing.

RQ5: Can automated image classification using deep learning support practical wildfire monitoring and early warning systems?

The results suggest that deep learning-based image classification systems can provide a reliable foundation for automated wildfire monitoring. The achieved accuracy and class-level performance indicate that such systems can support environmental monitoring and early warning applications. However, further testing under real-world conditions and integration with operational monitoring infrastructures would be necessary before deployment.

5.3 Comparison with Previous Studies

The findings obtained in this study are consistent with the growing body of research demonstrating the effectiveness of deep learning techniques for wildfire detection and environmental monitoring applications. Previous studies reviewed in Chapter 2 reported strong classification performance using Convolutional Neural Networks, transfer learning architectures, and hybrid deep learning approaches. Researchers such as Karaş et al. demonstrated that CNN-based models can achieve high wildfire detection accuracy using satellite imagery, while other studies highlighted the effectiveness of transfer learning architectures including ResNet, VGG, and EfficientNet for fire-related image classification tasks. The results obtained in the present study further support the conclusion that deep learning models are capable of learning meaningful visual representations associated with wildfire-related events.

A notable finding of this research is that the Custom CNN model outperformed the ResNet50 transfer learning architecture. While several previous studies have reported strong performance from transfer learning approaches, the results of this study suggest that task-specific architectures can achieve superior performance when sufficient domain-specific training data is available. The Custom CNN achieved an accuracy of 88.30%, whereas ResNet50 achieved 78.83%. This outcome indicates that pretrained feature representations are not always optimal for specialized environmental monitoring tasks. The findings therefore contribute to the ongoing discussion regarding the relative advantages of custom deep learning architectures and transfer learning approaches in computer vision applications.

The present study also extends previous research by providing a detailed evaluation using multiple performance metrics including accuracy, precision, recall, F1-score, confusion matrices, and classification reports. Many published studies focus primarily on overall classification accuracy, whereas this research provides a more comprehensive assessment of model behavior across individual classes. Furthermore, the study demonstrates the importance of preprocessing techniques, data augmentation strategies, and model-specific architectural choices in determining classification performance. Consequently, the findings contribute additional evidence supporting the application of deep learning techniques for automated wildfire monitoring and provide useful insights for future research in this domain.

5.4 Limitations

Although the proposed wildfire classification framework achieved promising results, several limitations should be acknowledged when interpreting the findings. First, the study utilized a single publicly available dataset obtained from Kaggle. While the dataset contained a large number of images representing Fire, Smoke, and Non-Fire categories, it may not fully capture the diversity of real-world wildfire conditions encountered across different geographical regions, weather environments, and monitoring systems. Consequently, the reported performance may not generalize perfectly to all operational scenarios.

A second limitation relates to the scope of model evaluation. The study focused on image classification performance using static images and did not investigate real-time wildfire detection using video streams, surveillance systems, or drone-based monitoring platforms. Real-world deployment environments often introduce additional challenges including image noise, motion blur, varying camera angles, and changing environmental conditions. Therefore, further testing is required before the proposed framework can be considered suitable for operational deployment.

Another limitation concerns explainability and model interpretability. Although the study discussed the importance of Explainable Artificial Intelligence techniques such as Grad-CAM and SHAP, these methods were not implemented within the current experimental framework. As a result, the internal decision-making processes of the developed models were not examined. Future research incorporating explainability techniques would provide additional confidence regarding the reliability and trustworthiness of model predictions. Finally, only one transfer learning architecture (ResNet50) was evaluated. Additional comparisons involving architectures such as EfficientNet, DenseNet, MobileNet, and Vision Transformers may provide further insights into optimal model selection for wildfire image classification tasks.

5.5 Future Work

The results obtained in this study demonstrate the potential of deep learning techniques for automated wildfire image classification; however, several opportunities exist for further enhancement and expansion of the proposed framework. One important direction for future research is the evaluation of additional deep learning architectures beyond the models investigated in this study. Advanced architectures such as EfficientNet, DenseNet, MobileNet, ConvNeXt, and Vision Transformers (ViT) have demonstrated strong performance in computer vision applications and may provide improved classification accuracy and computational efficiency for wildfire detection tasks. Comparative evaluations involving a wider range of architectures would provide deeper insights into optimal model selection strategies for environmental monitoring applications.

Another promising area for future investigation is the incorporation of Explainable Artificial Intelligence (XAI) techniques. Methods such as Grad-CAM and SHAP can provide visual explanations of model predictions and help researchers understand which image regions contribute most strongly to classification decisions. The integration of explainability techniques would improve transparency, facilitate model validation, and increase confidence in deployment scenarios where decision reliability is critical. Furthermore, explainability analysis may help identify hidden biases, improve error analysis, and support the development of more trustworthy wildfire detection systems.

Future work may also focus on expanding the practical applicability of the proposed framework through real-time deployment and multimodal monitoring approaches. The integration of deep learning models with surveillance cameras, drone platforms, satellite imagery systems, and Internet of Things (IoT) sensors could support continuous wildfire monitoring and early warning applications. Additional research involving video-based fire detection, temporal analysis, edge computing deployment, and geographically diverse datasets would further strengthen system robustness and operational readiness. These enhancements would contribute toward the development of intelligent wildfire monitoring systems capable of supporting environmental protection, disaster management, and public safety initiatives.

6 Conclusion

This study investigated the application of deep learning techniques for automated wildfire image classification using a publicly available dataset containing Fire, Smoke, and Non-Fire images. The research aimed to evaluate the effectiveness of a Custom Convolutional Neural Network and a ResNet50 transfer learning model for accurately classifying wildfire-related images. A comprehensive experimental framework was developed involving dataset preprocessing, image augmentation, model training, performance evaluation, and comparative analysis. The results demonstrated that both deep learning approaches were capable of identifying wildfire-related visual patterns; however, the Custom CNN consistently outperformed the transfer learning model across the evaluated metrics.

The experimental evaluation showed that the Custom CNN achieved a test accuracy of 88.30%, whereas the ResNet50 transfer learning model achieved an accuracy of 78.83%. The classification report further confirmed the effectiveness of the Custom CNN, with strong precision, recall, and F1-score values across all classes. Confusion matrix analysis demonstrated that the majority of images were correctly classified, indicating the ability of the proposed architecture to learn meaningful visual representations associated with wildfire-related conditions. These findings suggest that task-specific deep learning architectures can provide highly effective solutions for wildfire image classification when sufficient domain-specific training data is available.

The study also highlighted the importance of preprocessing techniques, data augmentation strategies, and systematic evaluation methodologies in developing reliable deep learning systems. Through comparative analysis, the research demonstrated that model selection should consider both architectural complexity and application-specific requirements. While transfer learning remains a valuable strategy in many computer vision tasks, the findings indicate that a carefully designed Custom CNN can outperform a pretrained architecture for specialized environmental monitoring applications.

Overall, the research successfully achieved its objectives and answered the proposed research questions. The findings confirm that deep learning-based image classification systems can provide a reliable foundation for automated wildfire monitoring and early warning applications. Although certain limitations remain, including dataset scope, explainability considerations, and deployment constraints, the results provide strong evidence supporting the continued exploration of artificial intelligence technologies for environmental protection and disaster management. Future research incorporating advanced architectures, explainable AI techniques, and real-time deployment strategies can further enhance the effectiveness and practical applicability of automated wildfire detection systems.

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